



COS40007 Artificial Intelligence Engineering

Portfolio Assessment 1: Exploratory Data Analysis (EDA) & Data Preparation

Due: 18/03/2026 23:59 PM on Canvas

AIM

This task aims to demonstrate your familiarity with Python programming, your favourite IDE, engineering datasets and programming machine learning with Python. Conduct a comprehensive Exploratory Data Analysis (EDA) on a dataset relevant to your engineering discipline. This assessment builds the foundation for all subsequent modelling tasks in this course.

Task 1: Dataset Loading & Initial Exploration

Load your dataset and perform initial exploration using pandas.

Deliverables:

- Code showing data loading
- Dataset shape (rows, columns)
- First 5 rows display
- Data types of all columns
- Basic statistical summary
- Initial missing value check

Task 2: Data Quality Assessment

Systematically identify and document data quality issues, such as missing values, duplicate values, and outliers.

Deliverables:

- Missing values table with percentages
- Missing values heatmap
- Duplicate count and examples (if any)
- Boxplots for all numerical features
- Initial observations about data quality

Task 3: Target Variable Preparation

1.1 For Numerical Target Variables:

- Define 3 to 5 meaningful class labels by discretising your numerical target variable.
- Clearly justify how class boundaries were selected:
 - Equal-width binning
 - Equal-frequency binning
 - Engineering threshold standards



- Ensure the resulting classes are **reasonably balanced**.
- Show class distribution (e.g., bar plot or value counts).

1.2 For Categorical Target Variables:

If your target is already categorical, verify class distribution. If a severe imbalance (>80% in one class) exists, document this challenge and note how it will affect modelling.

Deliverables:

- Target variable name and data type
- Problem type determination (regression/classification)
- Target distribution visualisation
- Summary statistics
- Initial observations about the target

Task 4: Univariate Analysis

Analyse each predictor variable individually.

Deliverables:

- List of all predictors
- Distribution plots for numerical predictors (at least 6)
- Summary statistics table for all predictors
- Observations about distributions (normal, skewed, etc.)

Task 5: Multivariate Analysis

Analyse relationships between variables.

Deliverables:

- Correlation heatmap
- Top 3 features correlated with the target (with coefficients)
- Identification of multicollinearity (high correlations between predictors)
- Pairplot of key features
- Observations about relationships and patterns

Task 6: Engineering Interpretation

Connect your statistical findings to engineering domain knowledge.

Answer the following questions or think of your own relevant questions:

1. What do the correlations mean in engineering terms?
2. Are there any unexpected patterns from an engineering perspective?
3. What physical/process relationships do the data reflect?
4. Based on your engineering knowledge, are there any missing features that would be important?



Task 7: Challenges Encountered

Describe any difficulties with data cleaning, outliers, etc.

Task 8: Responsible AI Reflection Questions

- How might poor data quality affect model reliability?
- What ethical or bias issues might exist in this dataset?

Task 9: Summary Report

Compile all findings into a structured report using the template provided on Canvas.

Generative AI Use Declaration

For this portfolio assessment, you may use Generative AI (e.g., ChatGPT, GitHub Copilot, Claude) for:

Permitted Uses:

- Generating code comments and documentation
- Debugging assistance
- Explaining error messages
- Suggesting features of engineering ideas
- Improving code readability

Required Declaration:

Include the following statement at the beginning of your portfolio submission:

"I used [GenAI tool name] to assist with [specific tasks, e.g., commenting code, debugging, suggesting feature combinations]. All final code, interpretations, and conclusions are my own work after verification."

Failure to declare the use of GenAI will result in a 10-point deduction.

Submission Requirements

Format: Single Jupyter notebook/Python file pushed to GitHub Classroom and a PDF report containing:

- GenAI declaration at the beginning
- All code cells with outputs
- Visualisations (properly labelled)
- Markdown cells explaining your reasoning

File Naming:

- 1 Portfolio_Assessment_1_[Studio_Number]_[YourName].pdf
- 2 GitHub Classroom link to push your code: <https://classroom.github.com/a/775q4tU1>